

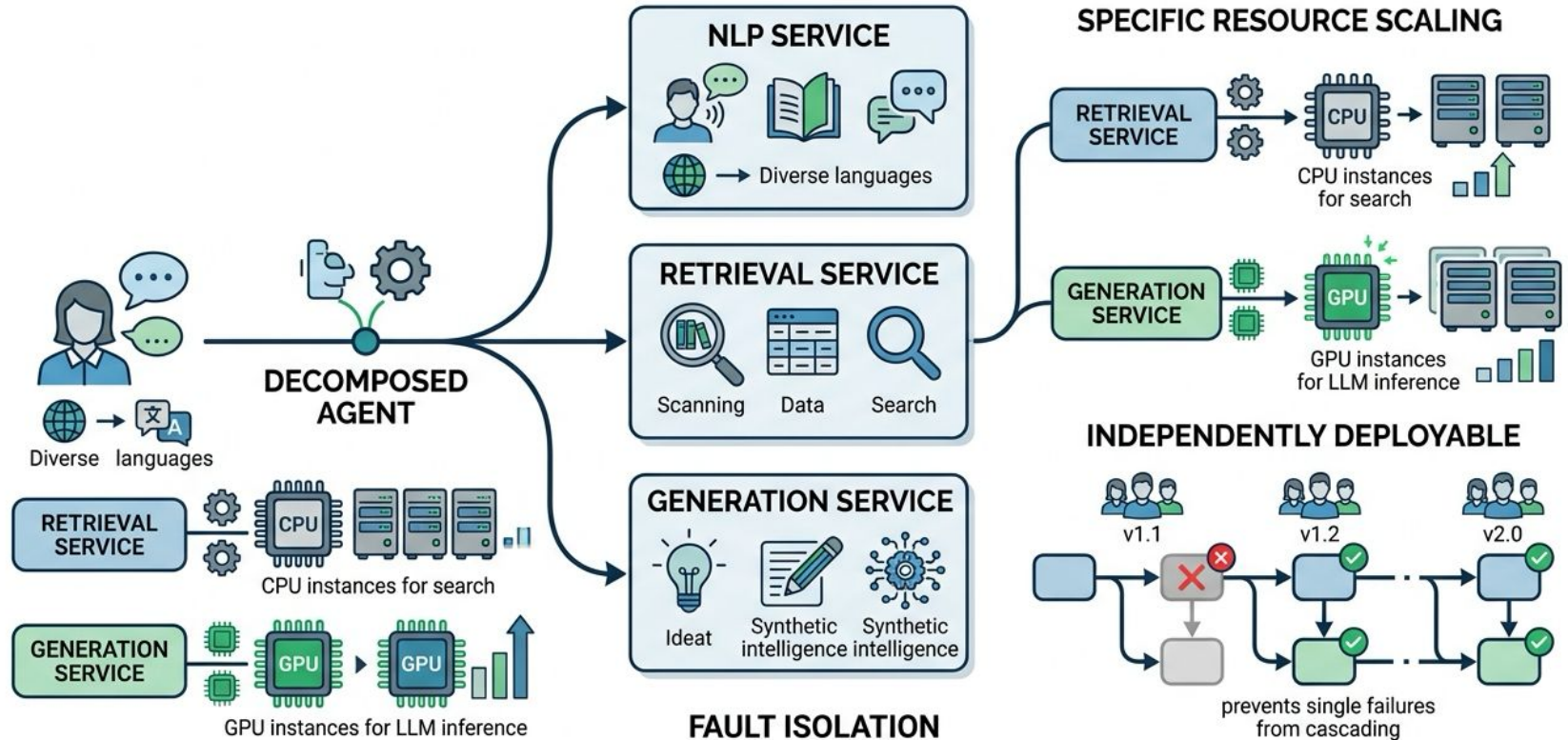


U.S. General Services
Administration

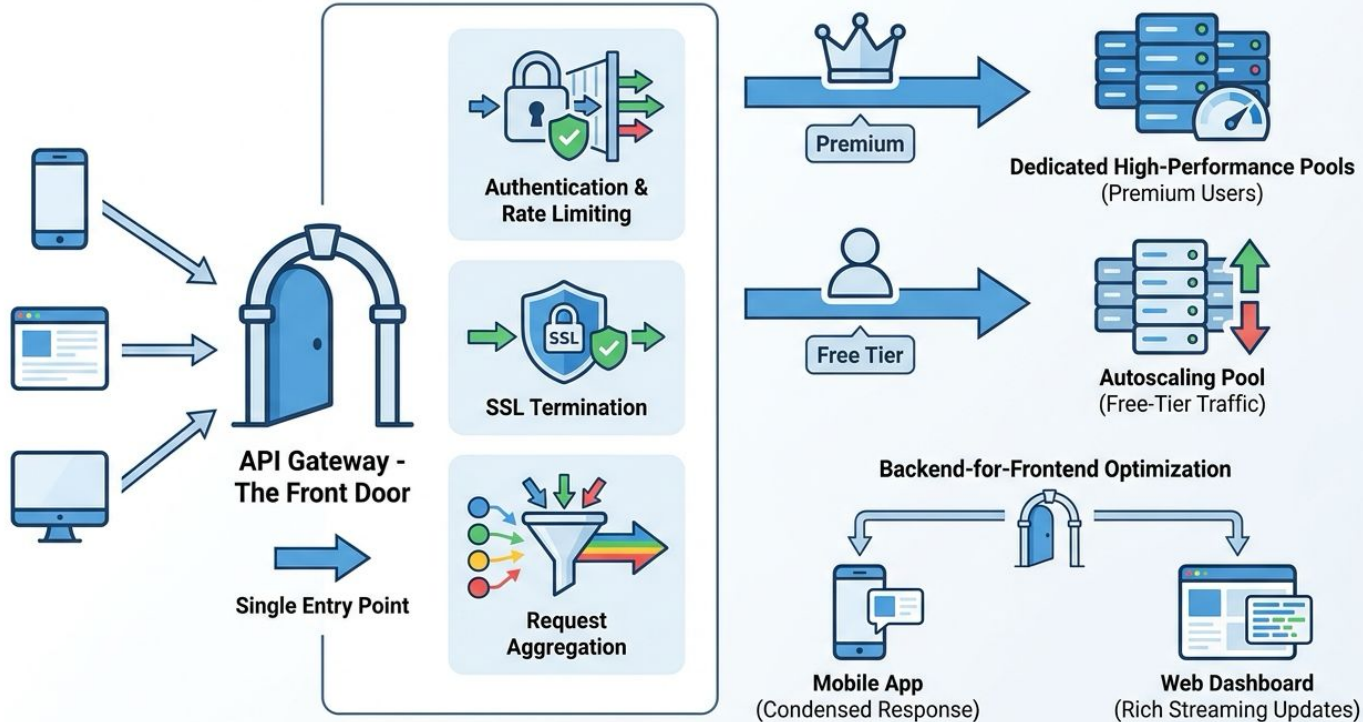
Chapter 4.2: Deployment and Scaling

US General Service Administration
AI Community of Practice - March 2026

Microservices Architecture for Agent Systems

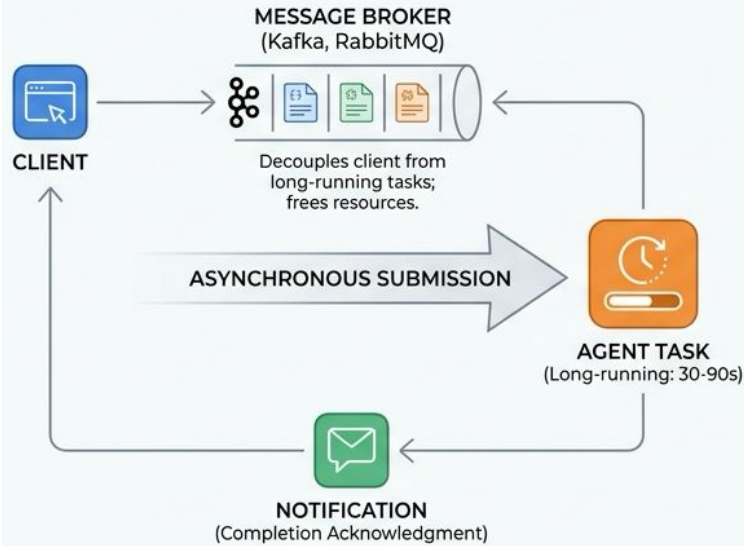


API Gateway -- The Front Door

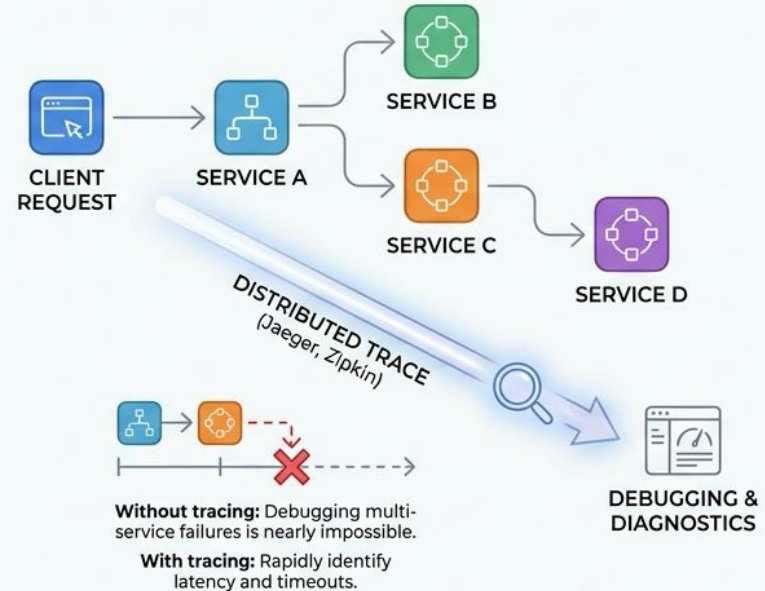


Event-Driven Communication and Tracing

EVENT-DRIVEN COMMUNICATION (ASYNCHRONOUS WORKFLOWS)

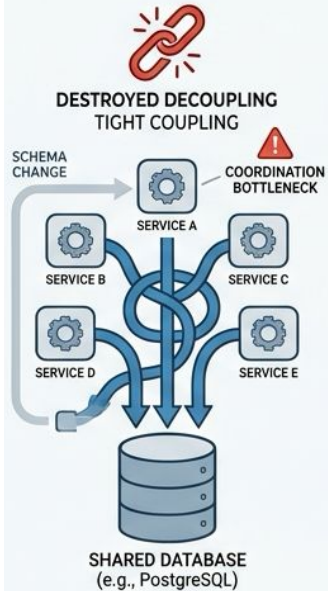


DISTRIBUTED TRACING (CROSS-SERVICE VISIBILITY)



Microservices Anti-Patterns

DISTRIBUTED MONOLITH



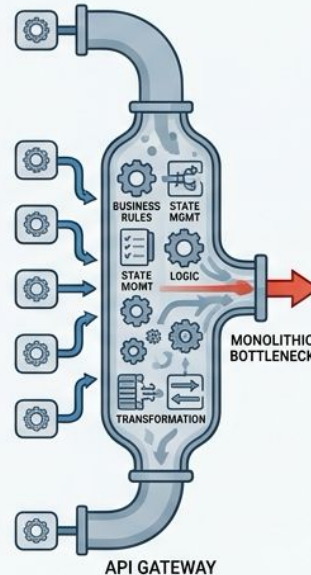
Operational Complexity,
No Decoupling.

OVER-FRAGMENTATION



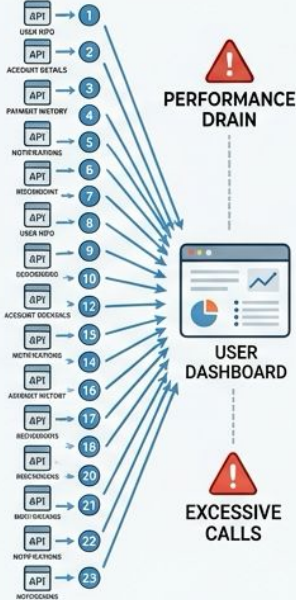
Architecture Requires 4K Monitor.
Excessive Coordination.

SMART GATEWAY PIPES



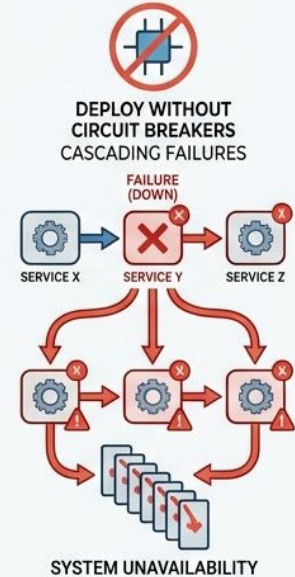
Gateway Implements Complex Logic.
Single Point of Failure.

CHATTY APIs



23 Calls for One Dashboard.
High Latency.

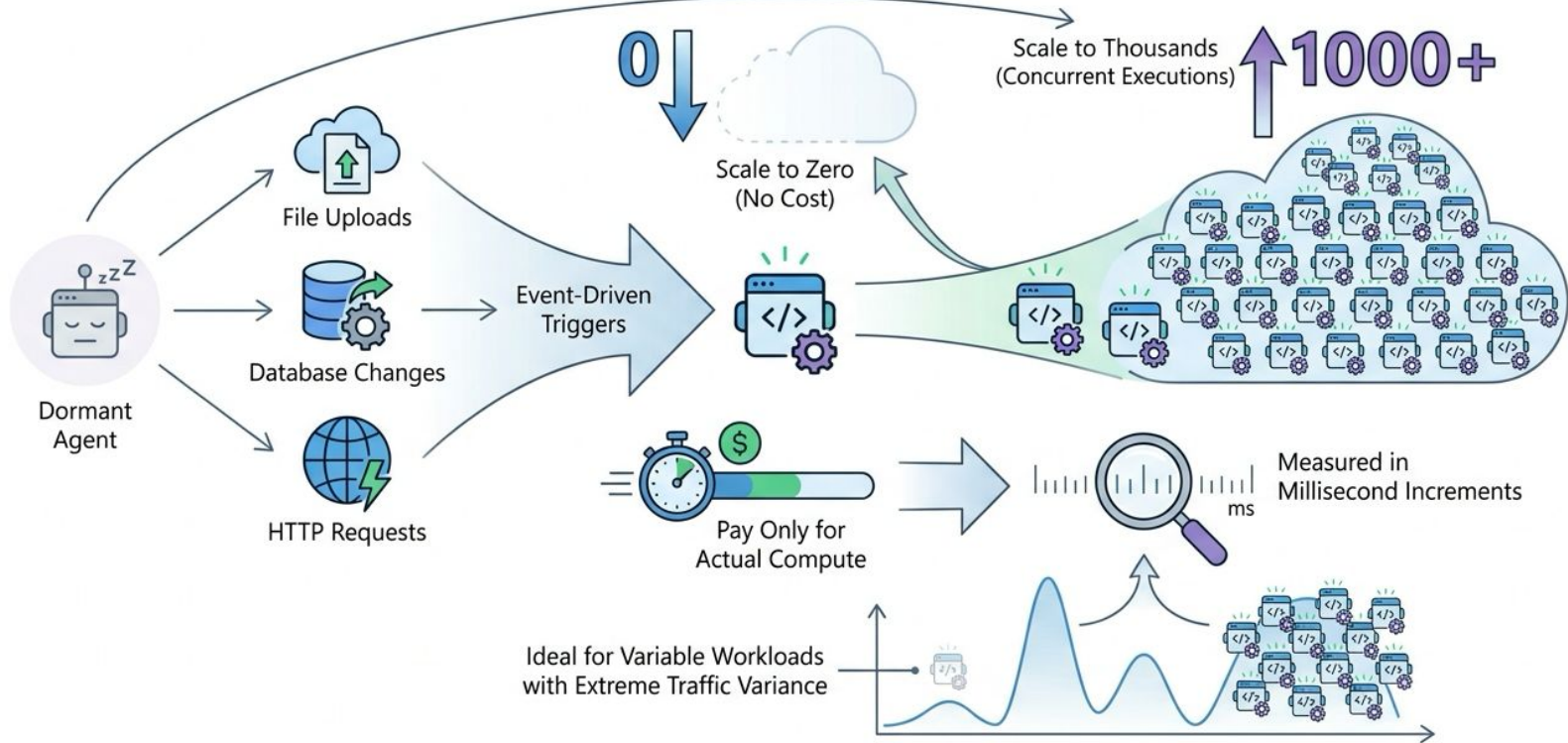
NO CIRCUIT BREAKERS



Transient Failure Cascades.
Total System Unavailability.

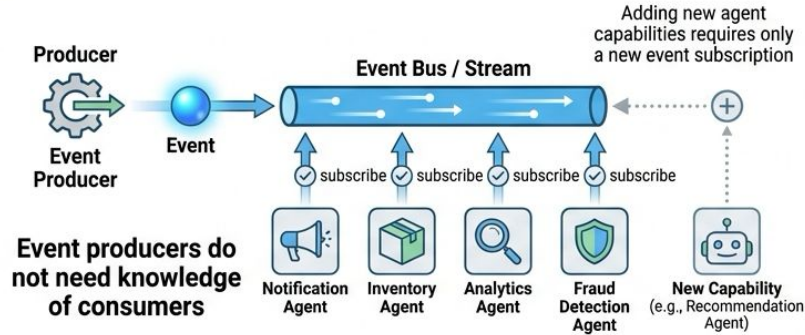
🕒 KEEP IT SIMPLE: GATEWAY LOGIC (AUTHENTICATE, RATE LIMIT, ROUTE). IMPLEMENT CIRCUIT BREAKERS. ⚠️ AVOID SHARED DATABASES.

Serverless Agents

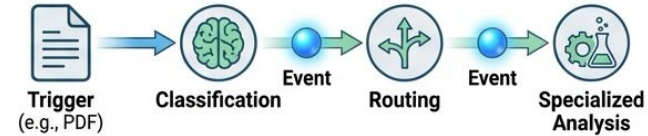


Serverless Design Patterns and Decoupling

DECOUPLED EVENT-DRIVEN ARCHITECTURE

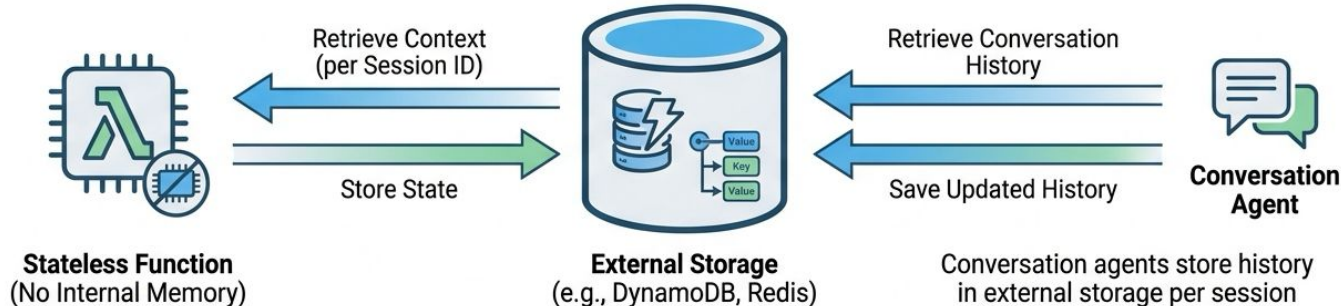


EVENT CHAINS REPLACE ORCHESTRATORS



Event chains replace central orchestrators for pipeline workflows

STATELESS DESIGN & EXTERNAL CONTEXT

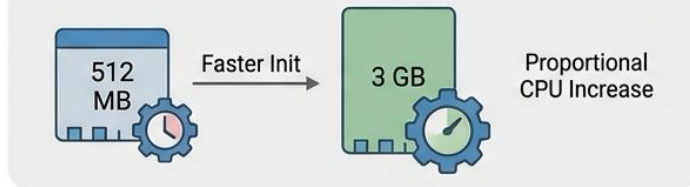


Serverless Pitfalls and Cold Start Reality

COLD START REALITY



Memory & CPU



VPC Configuration

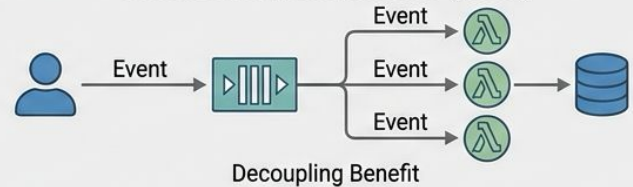


SERVERLESS PITFALLS

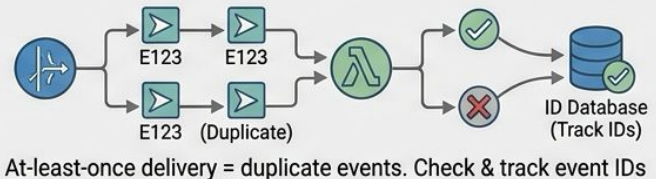
Synchronous Patterns



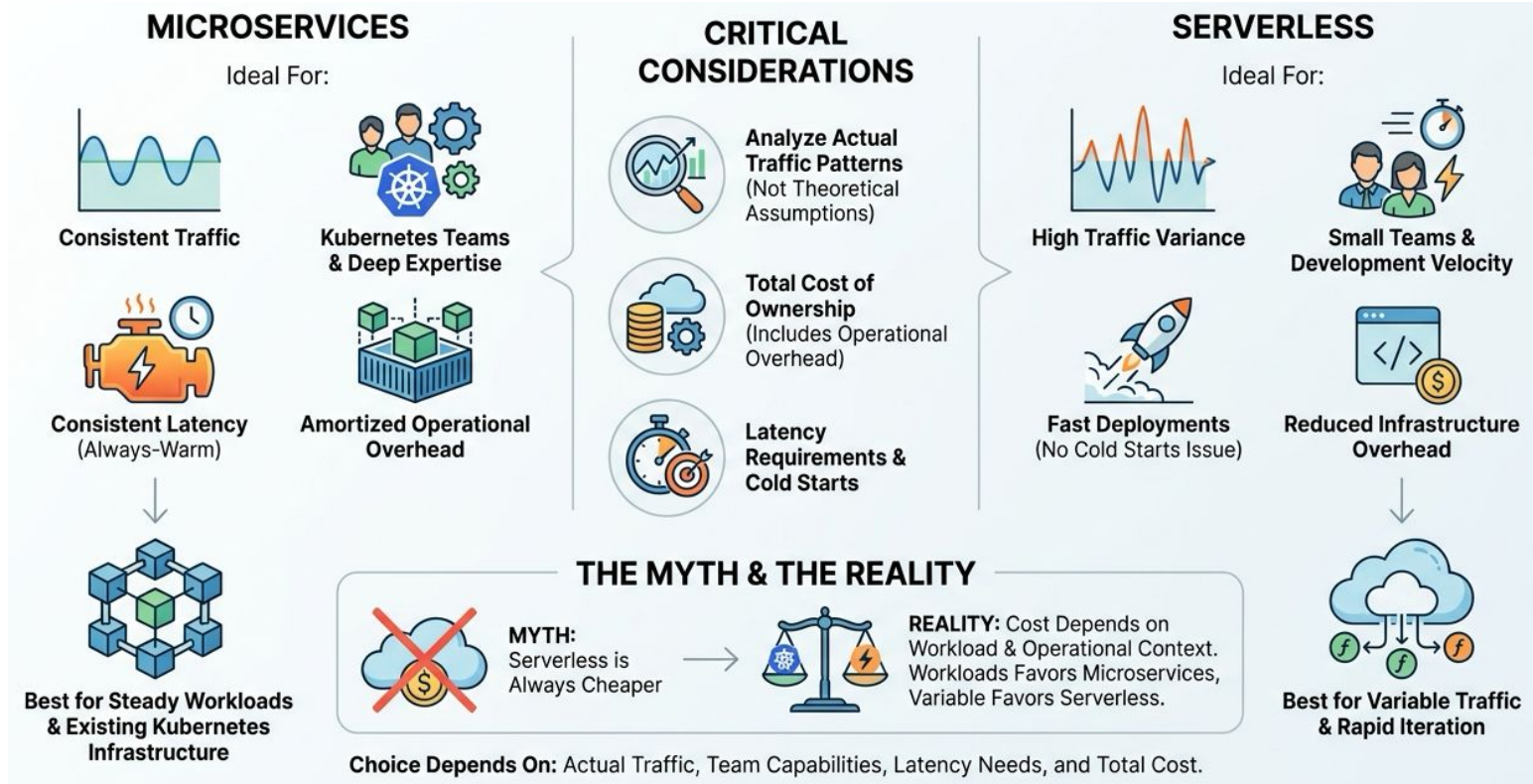
Event-Driven Architecture (Ideal)



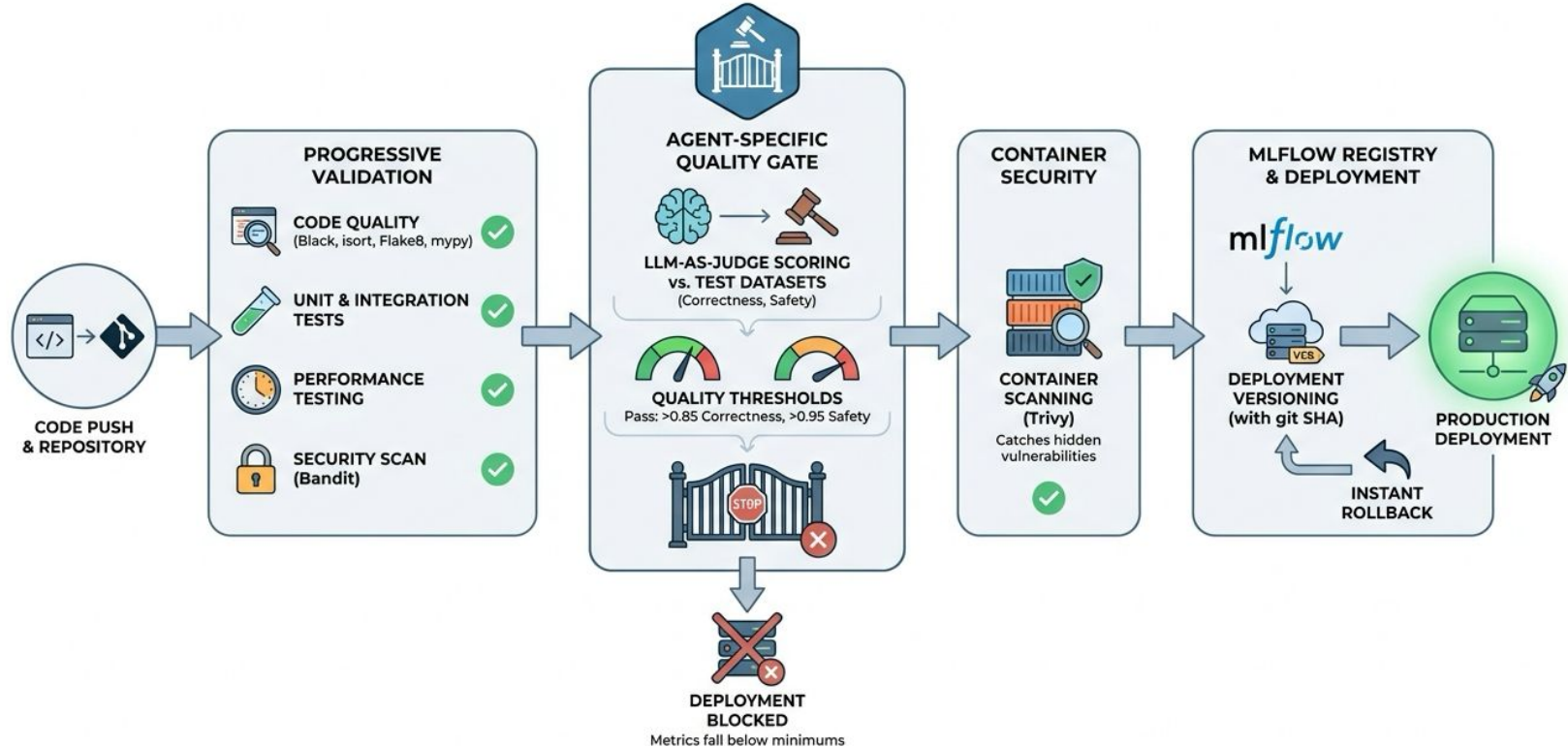
Implement Idempotency



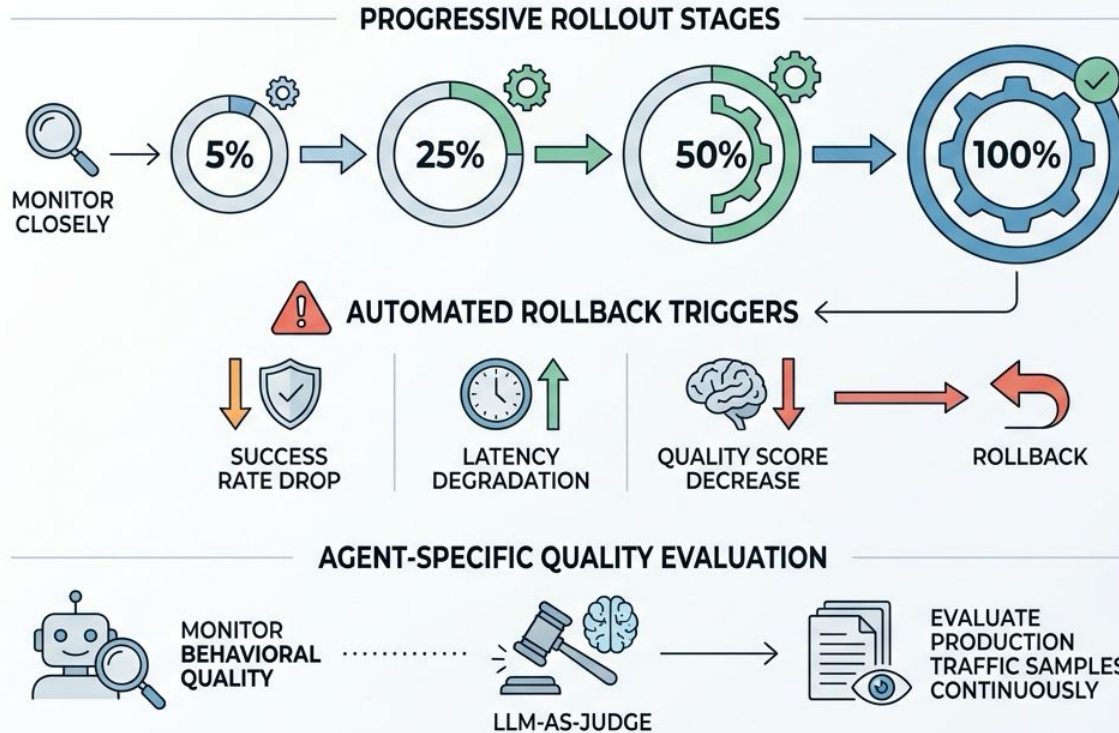
Microservices vs. Serverless -- Choosing Wisely



CI/CD Quality Gates for Agent Systems



Canary Deployments



Performance Bottlenecks

